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Improving the outcomes of anticoagulation: an evaluation of home follow-up of warfarin initiation

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Objectives. A number of studies have reported that the risk of bleeding associated with warfarin is highest early in the course of therapy. This study examined the effect of a programme focused on the transition of newly anticoagulated patients from hospital to the community.

Design. Open-label randomized controlled trial.

Setting. Home-based follow-up of patients discharged from acute care hospital in southern Tasmania, Australia.

Subjects. A total of 128 patients initiated on warfarin in hospital and subsequently discharged to general practitioner (GP) care were enrolled in the study. Sixty were randomized to home monitoring (HM) and 68 received usual care (UC).

Interventions. HM patients received a home-visit by the project pharmacist and point-of-care international normalized ratio (INR) testing on alternate days on 4 occasions, with the initial visit two days after discharge. The UC group was solely managed by the GP and only received a visit 8 days after discharge to determine anticoagulant control.

Results. At discharge, 42% of the HM group and 45% of the UC group had a therapeutic INR. At day 8, 67% of the HM patients had a therapeutic INR, compared with 42% of UC patients ($P < 0.002$). In addition, 26% of UC patients had a high INR, compared with only 4% of HM patients. Bleeding events were assessed 3 months after discharge and occurred in 15% of HM patients, compared with 36% of the UC group ($P < 0.01$).

Conclusions. This programme improved the initiation of warfarin therapy and resulted in a significant decrease in haemorrhagic complications in the first 3 months of therapy.

Keywords: anticoagulation, education, haemorrhage, INR monitoring, warfarin.

Introduction

Whilst being a very valuable drug, warfarin is also recognized as a high-risk drug for adverse events. Adverse events from warfarin use in Australia were estimated to cost over $100 million per annum in direct hospital costs alone, in 1992 [1].

A number of studies have reported that the risk of bleeding associated with warfarin is highest early in the course of therapy [2–7]. In one of these studies, the frequency of major bleeding decreased from 3.0% during the first month of outpatient warfarin therapy to 0.8% per month during the rest of the first year of therapy and to 0.3% per month thereafter [7].

Patients discharged from hospital are at high-risk of experiencing adverse events [8, 9]. Significant alterations to a patient's medication may occur in hospital. Patients may not take dosages appropriately after discharge and may not understand instructions appropriately. The initiation of warfarin is traditionally difficult despite the presence of a number of management protocols [10–12]. The trend to early discharge of patients with embolic disorders after the initiation of warfarin makes the transfer of information to general practitioners (GPs) very important.

The overall aim was to reduce complications associated with anticoagulants. In particular, the objectives of this study were to implement and evaluate a programme intended to promote the smooth transition of patient care from hospital to the community by incorporating home follow-up and point-of-care (POC) monitoring of patients commenced on warfarin.

Materials and methods

Patients were recruited from the Royal Hobart Hospital (RHH), a 450-bed acute care teaching hospital and the only major public hospital in the southern region of Tasmania (serving a population of approximately 230 000). Inpatients initiated on warfarin at the RHH between February 2002 and June 2003 were prospectively identified by the project pharmacist (SLJ) and ward pharmacists. Exclusions comprised patients who were not being discharged home in Southern Tasmania, who had dementia and were unable to answer basic questions about their therapy or who were entering a hospital in the home programme. Patients who were having warfarin re-initiated after surgery were also excluded.

Patients who provided informed consent were allocated to either an intervention (home monitoring; HM) or control (usual care; UC) group, using a computer-generated list of random numbers. Baseline demographic data, indication for anticoagulation, current medications and presence of contraindications to warfarin were assessed after consent for enrolment had been given. All patients received regular medical, nursing and pharmacy care during their hospitalization.

All GPs were sent a personalized information letter when their patient was discharged, indicating the group that the patient was enrolled to and what follow-up they would receive. Instructions in the letter to GPs caring for UC patients, indicated that this project was not responsible for the monitoring of their patients and patients would receive a visit from the project pharmacist 8 days after discharge to determine anticoagulant control. GPs caring for UC patients were telephoned on day 8 if an adverse trend was identified or if the international normalized ratio (INR) was out of the therapeutic range.

HM patients received a home-visit by the project pharmacist on alternate days on four occasions, with an initial visit 2 days after discharge from hospital. The project pharmacist tested the INR (CoaguChek S, Roche Diagnostics, Australia) and discussed a number of educational points regarding anticoagulant therapy, such as goals, possible adverse effects and interacting medications with warfarin. The pharmacist used a standard warfarin educational booklet as the basis of the counselling and also developed a double-sided A4 warfarin document to assist in education purposes (copies available from the authors). Patients' GPs were telephoned with each INR result during the four visits and subsequent dosage changes, if considered necessary, were discussed and implemented.

Details of conventional laboratory INR tests taken by the GP up to the day 8 visits (patient and/or GP reported) were also recorded. All patients were interviewed at 90 days after discharge to assess the type and frequency of anticoagulant-related complications. A systematic step-wise approach to assess the embolic and bleeding events (major and minor) was adapted from Heidinger *et al.* [13]. Medical record notes were examined for all patients readmitted during the study period to assess outcomes. The event rates were assessed through a combination of self-reported events and medical record notes.

Primary outcomes were (i) the achievement of a therapeutic INR value, as defined by the American College of Chest Physicians [14], on day 8 after discharge; (ii) total, major, minor bleeding complications; and (iii) readmissions to hospital due to anticoagulant-related complications within 90 days of initial discharge. Bleeding was defined as major if it was clinically overt and associated with either: a decrease in the haemoglobin level of at least 2 g dL^{-1} or the need for a transfusion of two or more units of red cells, if it was retroperitoneal or intracranial, if it warranted the permanent discontinuation of warfarin or if it required hospital admission. All other bleeding complications were assessed as minor. The medical records of all patients who reported symptoms of embolic complications [transient ischaemic attack (TIA), stroke, acute myocardial infarction (AMI), complications of deep vein thrombosis (DVT)] were examined and GPs contacted if necessary to confirm these events. Secondary outcomes were unplanned readmissions from any cause, death (cause determined from death certificates; no autopsies were performed), and proportion of patients remaining on anticoagulant therapy. Follow-up was complete through patient interview and medical record review for all patients.

Rates of outcomes were expressed as the number and percentage of events with 95% CI. Differences between groups were compared with Mann–Whitney or Kruskal–Wallis tests. Changes within groups for numerical variables were compared by Wilcoxon-signed rank tests. Categorical variables were compared by chi-square tests.

Based on previous studies showing approximately 30% of patients commenced on warfarin experience a bleeding complication within 3 months [2, 4] and given that the aim of the intervention programme was to reduce this figure to below 10%, it was estimated that 72 patients were needed per group (at a power of 80% and $P = 0.05$).

The RHH Research Advisory Committee and the Southern Tasmania Health & Medical Human Research Ethics Committee had approved the project.

Results

A total of 131 patients were identified as eligible for inclusion (Fig. 1). Table 1 summarizes the clinical and demographic features of the 128 patients enrolled in the study. The two groups were well matched with regard to baseline demographics. There was a significantly higher incidence of previous myocardial infarction in the HM group compared with the UC group. One patient (HM) was excluded after the day 8 follow-up, due to the changing of therapy from warfarin to low-molecular weight heparin on advice from the GP on day 8. Therefore, outcomes at 90 days were assessed for 127 patients.

Table 2 displays the quality of anticoagulation for all patients on discharge and at day 8 after discharge from the hospital. There were no significant differences between the groups in regard to anticoagulant control at discharge. However, there was a significant difference in anticoagulant control at day 8 after discharge between the HM and UC groups (67 and 41% therapeutic, respectively, $P < 0.01$). The median INR at discharge in the HM and UC groups was 2.0 (1.0–3.6) and 2.2 (1.0–4.0), respectively ($U = 1742.5$, $P = 0.55$). The median INR at day 8 after discharge in the HM and UC groups was 2.4 (1.3–5.8) and 2.1 (1.0–7.6), respectively ($U = 1602.5$, $P = 0.84$). The median days of warfarin dosing before discharge for all patients was 6 (1–45) and did not differ between the groups. Forty-four per cent of all anticoagulant initiations followed the RHH's anticoagulation initiation protocol, 40% in the intervention group compared with 47% in the control group ($\chi^2 = 0.62$, $P = 0.43$). Table 3 displays the level of anticoagulation for the HM group at each visit after discharge, with two-thirds of patients in the intervention group having a therapeutic INR at day 8 after discharge.

There was a trend towards significance for patients in the UC group with a sub-therapeutic INR at discharge to have a poorer outcome at day 8 compared with therapeutic and supra-therapeutic patients in the same group ($\chi^2 = 6.8$, $P = 0.14$). Patients in the HM group who had sub-therapeutic INRs at discharge were significantly more likely than the corresponding patients in the UC group to have an INR in the therapeutic range at day 8 (62 and 37%, respectively; $\chi^2 = 11.9$, $P = 0.003$). Importantly, 37% of patients who were discharged sub-therapeutic in the UC group had a supra-therapeutic INR at day 8 compared with none in the HM group.

The median duration of pharmacist visits to the HM group after discharge was 24 min across the four visit days. After the day 8 visit by the pharmacist to UC patients, 10% had a subsequent dose increase and 13% had a dose decrease. The median number of conventional laboratory INR tests performed during the 8 days following hospital discharge was 0 (0–2) in the HM group and 2 (0–6) in the UC group.

Only 24 (37%) UC patients were using their standard issue warfarin booklet to record their INR results at day 8 compared with 100% of HM patients ($\chi^2 = 55.6$, $P < 0.0001$). At 90 days after initial discharge, 20 (31%) UC patients compared with 28 (51%) HM patients were using their warfarin booklets to record their INR results ($\chi^2 = 4.75$, $P = 0.02$).

Table 4 displays the number and rates of adverse events occurring up to 90 days after discharge for all patients. There was a significant difference in total, major and minor bleeding events between the HM and UC groups. All unplanned cessations of warfarin assessed at 90 days after discharge

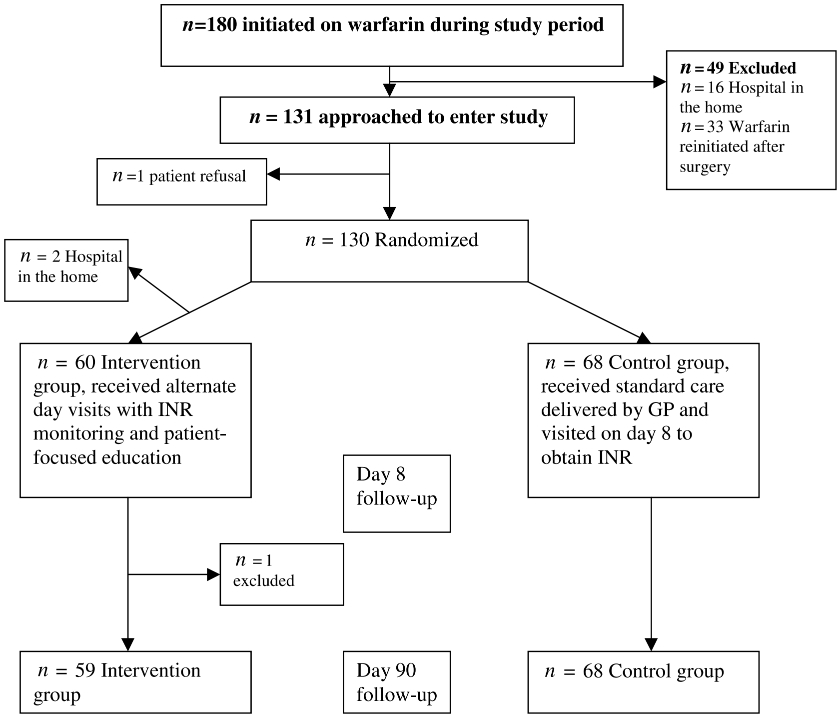

Fig. 1 Recruitment flowchart.

Table 1 Baseline characteristics of trial participants

	Home monitoring ($n = 60$)	Usual care ($n = 68$)
Age in years – median (range)	70 (19–94)	72.5 (20–91)
Female (%)	47	47
Cardiovascular admission (%)	65	65
Hypertension (%)	53	47
Diabetes (%)	13	19
Previous AMI (%)	18*	6*
Congestive heart failure (CHF) (%)	22	19
Malignancy (%)	7	6
Contraindications to warfarin (%)	20	10
Previous warfarin use (%)	13	13
Initial hospital stay median (range)	8 (3–72)	8 (1–65)
No. comorbidities median (range)	4 (1–12)	4 (1–9)
No. regular medications median (range)	6 (1–12)	6 (1–15)
Reason for warfarin therapy		
Atrial fibrillation (%)	45	46
DVT or pulmonary embolism (%)	32	29
Valve replacement (%)	18	18
Mural thrombus (%)	5	7

*$P < 0.05$.

occurred as a result of death or complications of anticoagulant therapy. There did not appear to be any significant correlations between anticoagulant control at day 8 (supra-, sub- or therapeutic) and subsequent bleeding (total, major or minor) complications for all patients, for the HM group or UC groups. However, 25% (four events) of the patients in the UC group who had a supra-therapeutic INR at day 8 had a subsequent embolic event during the 90 days after discharge, compared with one event each in the sub- and therapeutic patients ($\chi^2 = 5.97$, $P < 0.05$). There was no significant correlation between anticoagulant control at day 8 and embolic outcomes for the HM group.

Table 2 Anticoagulant control at discharge and at day 8 after discharge

	Discharge		Day 8 after discharge[a]	
	Home monitoring ($n = 59^{b}$) (%)	Usual care ($n = 66^{b}$) (%)	Home monitoring ($n = 52^{c}$) (%)	Usual care ($n = 63^{c}$) (%)
Sub-therapeutic	49	47	29	33
Therapeutic	42	45	67	41
Supra-therapeutic	9	8	4	26

[a]$P < 0.01$ for differences between home monitoring and usual care groups.
[b]Some patients did not have an international normalized ratio obtained on the day of discharge.
[c]Some patients were readmitted before the day 8 home visit.

Table 3 Anticoagulant control in the home monitoring group by day of follow-up

	Day 2 ($n = 60$) (%)	Day 4 ($n = 60$) (%)	Day 6 ($n = 56$) (%)	Day 8 ($n = 52$) (%)
Sub-therapeutic	57	42	41	29
Therapeutic	38	40	46	67
Supra-therapeutic	5	18	13	4

Discussion

There is clear evidence that the trend to early discharge for hospital patients is putting strain on the primary care system. This study, which commenced in 2002, shows that 44% of all patients had a therapeutic INR at discharge, compared with 63% in a previous audit of anticoagulant therapy at the same hospital, completed in 1994 [15]. The median days of initiation of warfarin in the previous study was 8 days, compared with 6 in this study.

This study shows the actual practice of initiation of warfarin in a teaching hospital and subsequent discharge of patients to community care may result in poor outcomes. The combination of shorter periods of hospitalization and increasing usage of warfarin is placing stress on general practice health services to care for newly anticoagulated patients.

The majority of the study population represented those chronically ill older patients, who were initiated on warfarin for the treatment of chronic conditions. The two groups were well matched in baseline demographics, except for a significantly higher incidence of previous AMI in the HM group.

Table 4 Adverse outcomes occurring up to 90 days after discharge

	Home monitoring ($n = 59$)	Usual care ($n = 68$)	P-value
Total bleeding	15 (7–27)	36 (24–48)	0.009
Major bleeding[a]	2 (0–9)	10 (4–20)	0.05
Minor bleeding	15 (7–27)	34 (23–46)	0.01
Embolic complication[b]	9 (3–19)	10 (4–20)	0.73
Unplanned readmission	22 (12–35)	27 (17–39)	0.56
Unplanned cessation of warfarin	12 (5–23)	10 (4–20)	0.55
Readmission due to warfarin complication	3 (0–12)	8 (2–16)	0.32
Death[c]	7 (2–16)	8 (2–16)	0.90

[a]Usual care ($n = 7$). Recurrent haematuria, requiring cessation of warfarin; Massive epistaxis requiring hospital admission, INR 9.9; Recurrent gastrointestinal bleeding, warfarin ceased, hospitalized and transfused two units packed cells, INR on admission 2.6; massive epistaxis and gastrointestinal bleeding, requiring hospital admission, INR 8.3; severe bruising causing cessation of warfarin by GP; recurrent haematuria for investigation, INR on admission 2.2; gastrointestinal bleeding, requiring hospital admission, 4 units of packed cells given and vitamin K administered, INR on admission 4.7. Home monitoring ($n = 1$); recurrent gastrointestinal bleeding, transfused two units packed cells, requiring hospital admission.
[b]Usual care ($n = 7$). Myocardial infarction ($n = 3$); transient ischaemic attack ($n = 2$); admission to hospital with symptoms of embolic complication (DVT) with sub-therapeutic INR; direct current reversion cancelled due to sub-therapeutic INR and suspected formation of left atrial emboli on echocardiography. Home monitoring ($n = 5$); myocardial infarction ($n = 2$); unstable angina pectoris; stroke (embolic; diagnosed on computed tomography scan); admission to hospital with symptoms of embolic complication (DVT) with sub-therapeutic INR.
[c]Usual care ($n = 5$). Myocardial infarction ($n = 2$); Sepsis with bone marrow suppression and disseminated intravascular coagulation; cardiac dysrhythmia on background of congestive cardiac failure and atrial fibrillation; cardiac arrest on background of ischaemic heart disease. Home monitoring ($n = 4$); myocardial infarction ($n = 2$); thalamic tumour; cerebrovascular accident on background of arteriosclerosis and hypertension.

The authors are unaware of any lifestyle differences between the two groups that could have been responsible for this difference. However, if a number of baseline demographics are reported, the risk of random chance contributing to a significant difference in these variables increases. Elderly patients are not only at an increased risk of bleeding complications but are also have a higher benefit with anticoagulant therapy compared with younger populations [16, 17]. In fact, nearly half of the patients had warfarin initiated for stroke prophylaxis in atrial fibrillation. This group of patients are at highest risk of medication misadventure if not adequately monitored and educated.

This study examined the effect of POC monitoring of anticoagulant therapy, with patient-focused education by a pharmacist, amongst a population of newly initiated warfarin patients. We found that this comprehensive programme reduced the frequency of bleeding in patients randomly assigned to home monitoring at the start of anticoagulant therapy.

Less than adequate initiation of warfarin may have caused problems after discharge in both groups of patients. Despite the availability of initiation protocols in the hospital, the initiation of warfarin continues to be a problem, with only 44% of all warfarin initiations following the institutional protocol. The anticoagulant effect of any daily dose of warfarin in the initiation phase is generally seen after 2 days [18]; therefore, poor initiation of anticoagulation whilst in hospital may have an impact on postdischarge INRs.

The initiation of warfarin in hospital and adherence to the hospital's anticoagulation protocol were similar in the two groups of patients. Therefore, it is unlikely that differences in warfarin initiation between the control and intervention groups are responsible for different outcomes at day 8. The HM group had two-thirds of patients in the therapeutic range at day 8, which had increased from 42% at discharge. Conversely, the proportion of patients in the UC group with therapeutic INR at day 8 was less than the proportion therapeutic at discharge. The alarming aspect of the follow-up at day 8 in the UC group was the relatively high proportion of supra-therapeutic INRs. These supra-therapeutic INRs at day 8 frequently occurred in patients who had a sub-therapeutic INR at discharge, reinforcing the benefit of close monitoring of patients who are not therapeutic or stable.

This project is an example of a systems solution to improve the management of anticoagulation, which looks to improve the processes of care rather than individual behaviour. When processes of care are examined, a common root cause of medication errors occurs at the time when decisions about therapy are made [19, 20]. Failure to obtain sufficient information about the patient or about the pharmaceutical agent has contributed to medication errors [21]. The programme aimed at providing patients and GPs with information regarding dose decisions at the point-of-care.

Despite low rates of warfarin-related bleeding reported in randomized trials, [22–26] which included strict exclusion criteria, higher rates of major bleeding have been reported in many studies of warfarin used in clinical practice [2, 27, 28]. In this study, the sample was assembled with few exclusion criteria, and the bleeding rates in the UC group were similar to those observed in other clinical practice studies. The programme achieved a frequency of major bleeding that was similar to the lower rates reported in previous randomized trials of the efficacy of warfarin.

A study by Beyth *et al.* [29] evaluated an intervention that consisted of patient education, training to increase patient participation, self-monitoring of prothrombin time, and guideline-based management of warfarin dosing compared with normal care for 6 months. The cumulative incidence of major bleeding in the usual care group was similar to our study, 12% and the incidence of major bleeding was reduced to 5.6% in the intervention group. Importantly, they noted that after 6 months, major bleeding occurred with similar frequencies in the intervention and usual care groups.

It is difficult to attribute the reduced bleeding complications in the HM group to any single part of the intervention. The study was not designed to determine the relative importance of the components of the intervention. It is likely that the combination of improved monitoring postdischarge and a patient-focused education programme has resulted in a reduction in the number of bleeding complications. Adherence to anticoagulant treatment is enhanced by knowledge and understanding of the drug, its benefits and side-effects [30–32].

The impact that the trial had on the UC group is hard to measure, with nearly one-quarter of the UC group having a dose change after the day 8 visit by the pharmacist. It is likely that the visit by the project pharmacist on day 8 prevented adverse events in some of the patients visited and resulted in an underestimation of the effect of the intervention.

Data for anticoagulant control between day 8 and day 90 after discharge could not be obtained for patients. Therefore, the contribution of the intervention to anticoagulant control between these time intervals could not be assessed. However, a higher proportion of patients in the UC group with a supra-therapeutic INR at day 8 subsequently developed embolic complications assessed at 90 days after discharge. This may have reflected poor control at day 8 as a marker for erratic control up to 90 days after discharge, potentially contributing to an increased risk of embolic complications.

In conclusion, this programme improved the initiation of warfarin therapy and resulted in a significant decrease in hemorrhagic complications in the first 3 months of therapy. The widespread adoption of similar programmes should be promoted, to improve the transition between hospital and community-based care for patients commenced on warfarin.

Conflict of interest

No conflict of interest was declared.

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